

10

ABA patterns and the generative power of Nanosyntax

Daniar Kasenov

10.1 Introduction

In linguistics, a theoretical proposal is often evaluated based on its generative power: what kinds of languages does the proposal predict to be possible? How does the prediction correspond to the observed crosslinguistic data? This chapter is an exercise in such an evaluation. The theory under the microscope is Nanosyntax, a late-insertion, syntax-all-the-way-down theory of morphosyntax.

The relevant property for the current evaluation is the possibility of generating the so-called (pseudo-)ABA patterns. In contemporary morphosyntax, it is assumed that ABA patterns are impossible (Bobaljik 2012 onward). By ABA patterns I mean “morphological patterns in which, given some arrangement of the relevant forms in a structured sequence, the first and third [forms] may share some property ‘A’ only if the middle member shares that property as well. If the middle member is distinct from the first, then the third member of the sequence must also be distinct” (Bobaljik and Sauerland 2018: p.2). Despite the ever-growing body of evidence that *ABA restriction holds across many morphological domains (e.g. Caha 2009; Lander and Haegeman 2018; Middleton 2021; Sudo and Nevins 2022), problematic patterns for an across-the-board ban on ABA phenomena are found again and again, in a variety of ways (see Middleton 2021 for the notion of “pseudo-ABA” patterns of A–B–Ax type, in which the middle paradigm cell is a portmanteau form; also see Caha’s 2024 distinction between “base ABA” and “full ABA”). The goal is, then, to make sure that a morphological theory fits in the middle ground of capturing the *ABA generalization and not ruling out attested ABA-like patterns.

Patterns of the form A–B–Ax are both attested and analyzable in Nanosyntax using either backtracking (Caha 2017) or complex left branches (CLBs) (also known as Movement-Containing Trees; see Blix 2021). This chapter’s contribution is that CLBs, when combined with pointers, the Nanosyntactic tool to model suppletion, give rise to an ABA suppletion pattern where all three cells of the paradigm are portmanteaus, which would be a purely ABA suppletion pattern.

The result adds to the research arguing that Nanosyntactic pointers are powerful additions to the machinery: for example, Taraldsen (2019) explores the fact that pointers can be used to derive ABA in multidimensional paradigms. This work shows that ABA patterns in one-dimensional paradigms are also within the reach of Nanosyntax with pointers. To supplement the theoretical argument, I argue that such patterns are attested and provide evidence from a substandard Russian combination of analytic comparative/superlative forms with suppletive adjectives. I then discuss the inter-speaker variability with respect to the pattern, suggesting that the discussed ABA pattern is unstable, which may have to do with the learnability bias in the tendency for ABA patterns to be unattested (cf. Andersson 2018). Summing up, the chapter aims to add to the emerging body of evidence that the independently motivated mechanisms in Nanosyntax (in this case, pointers) seem to derive just enough ABA patterns to fit the observed typology (see Introduction to this volume).

The chapter proceeds as follows. In Section 10.2, I give a quick introduction to the basic premises of Nanosyntax as understood in this chapter, making sure to introduce every theoretical notion necessary for my argument. Section 10.3 presents the theoretical discussion—first, I show how a pseudo-ABA pattern of A–B–Ax sort can be derived via both backtracking and CLBs (Blix 2021), using Malayalam pronominal paradigms as an example (Middleton 2021), and then I show how a putative ABA suppletion pattern is derived, using the combination of CLBs and pointers. Section 10.4 presents evidence from Russian that the predicted pattern is attested: I discuss the unexpected interaction between analytic comparatives/superlatives and suppletive adjectives and then argue that the pattern can only be understood as an ABA distribution of suppletive roots, providing support for the prediction of the Nanosyntactic model that ABA patterns with three portmanteaus are, indeed, possible. Section 10.5 concludes and shortly discusses the large-scale implications of the argument presented in this chapter.

10.2 The version of Nanosyntax assumed

As stated in the introduction, this chapter examines the limits of the Nanosyntactic model. Nanosyntax (see Starke 2009; Baunaz and Lander 2018 for introduction) is a late-insertion, syntax-all-the-way-down theory of morphology. These descriptions mean that Nanosyntax views morphological units (affixes, stems, ‘words’) as having complex internal structure (syntax all the way down), which is mapped onto observed forms via insertion rules that map syntactic objects onto phonological strings (late insertion).

However, these are the properties that Nanosyntax shares with the most prominent contemporary theory of syntax–morphology interface, Distributed Morphology (DM). The difference between Nanosyntax and DM lies in the modularity of morphology: DM assumes a modular postsyntax with a wide array of postsyntactic

operations that render the underlying syntactic structure opaque. Although there are works arguing for the necessity of such operations (Arregi and Nevins 2012; Hewett 2023), Nanosyntax goes in another direction by deriving the morphological phenomena using syntactic machinery.

Nanosyntax follows the cartographic tradition in generative syntax in assuming a version of the One Feature–One Head principle of Kayne (2005), according to which syntactic terminals are not bundles of features but rather individual features themselves (see Boeckx 2014 for a forceful argument against the idea of bundles of features). It is rather uncontroversial, however, that most morphological exponents realize multiple syntactic features. To handle this fact, Nanosyntax employs Phrasal Lexicalization (insertion rules map whole syntactic phrases to exponents, not single terminals, like in DM). The idea is that exponence of multiple features (=heads) by a single affix is the result of those heads forming a constituent in the syntax. An immediate question then arises: how are syntactic feature-heads grouped into constituents that undergo insertion? The answer is that Nanosyntax achieves that using syntactic movement. To give substance to the overview, let me refer back to the discussion of Turkish and Latin in the introductory chapter. Recall that in Turkish, number and case features are realized separately, as shown in Table 10.1.

Table 10.1 Fragment of the Turkish nominal paradigm

	Sg	Pl
Nom	<i>kitab</i> book	<i>kitab-lar</i> book-PL
Acc	<i>kitab-ı</i> book-ACC	<i>kitab-lar-ı</i> book-PL-ACC
Dat	<i>kitab-a</i> book-DAT	<i>kitab-lar-a</i> book-PL-DAT

In Nanosyntax, the structure that corresponds to the form *kitab-lar-ı* ‘books-PL-ACC’ is provided in the syntactic structure in Figure 10.1, where the PL head and the ACC head are found in different constituents. The idea is that two steps of movement have occurred: first, NumP has moved to Spec, PlP and then, the resulting PlP has moved to Spec, AccP. Note that I am working with a very simplified internal structure for nominals for exposition purposes.

Now let’s get back to the Latin case where the number and case features are realized together, as shown in Table 10.2 (I abstract away from the topic of declension classes for ease of exposition; see Caha 2021 and Blix 2021 for configurational proposals about declension classes in Nanosyntax).

The fact that the PL and ACC features are exponed together invites an analysis where the two feature heads form a constituent. The structure for the form *puer-ōs* ‘boy-ACC.PL’ is provided in the syntactic tree in Figure 10.2 with the idea being that, again, two steps of movement have occurred: first, movement of NumP to Spec, PlP and then movement of the NumP from Spec, PlP to Spec, AccP.

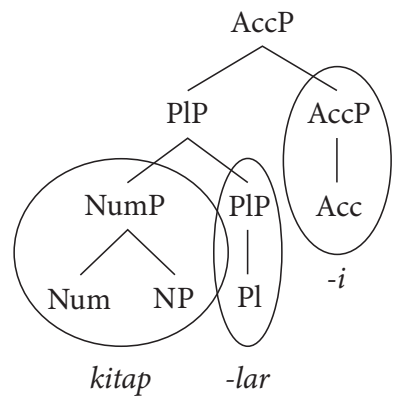


Figure 10.1 Structure for the Turkish form *kitap-lar-ı* ‘books-PL-ACC’

Table 10.2 A fragment of the Latin nominal paradigm

	Sg	Pl
Nom	<i>puer</i> boy.NOM.SG	<i>puer-ī</i> boy-NOM.PL
Acc	<i>puer-um</i> boy-ACC.SG	<i>puer-ōs</i> boy-ACC.PL
Dat	<i>puer-ō</i> boy-DAT.SG	<i>puer-īs</i> boy-DAT.PL

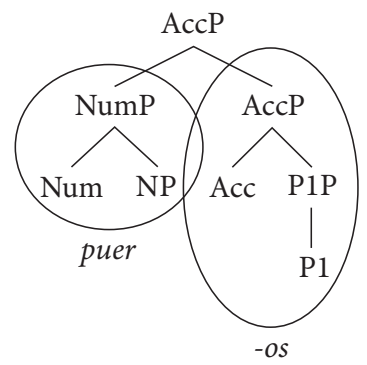


Figure 10.2 Structure for the form *puer-os* ‘boys-ACC.PL’

This exposition has introduced two main types of movement employed in Nanosyntactic derivations: Complement-to-Spec movement, which derives separate exponence of features, and Spec-to-Spec movement, which derives cumulative exponence of features. Nanosyntactic works employ the Lexicalization Algorithm (LA), which governs the order of these movements, provided in (5). The fourth step of the algorithm will be shown at work in the next section.

(1) Nanosyntactic LA

- a. Merge F to XP and lexicalize
- b. If (a) fails, move the constituent in Spec, XP to Spec, FP (Spec-to-Spec)
- c. If (b) fails, move XP to Spec, FP (Comp-to-Spec)
- d. If (c) fails, proceed to the next step in the previous cycle (backtracking)

The reliance of Nanosyntax on the LA is due to its commitment to the idea that lexicalization of syntactic structure proceeds bottom-up (Bobaljik 2000). The derivational nature of Nanosyntactic analyses invites a relevant topic of discussion, namely, the notion of Cyclic Override (Starke 2009). Cyclic Override means that successful lexicalizations override previous lexicalizations. Let us return to the Latin example. Assuming the containment of the accusative case in the dative case (Caha 2009), the lexicalized structure for the dative *puer-īs* ‘boys.DAT’ is given below in Figure 10.3.

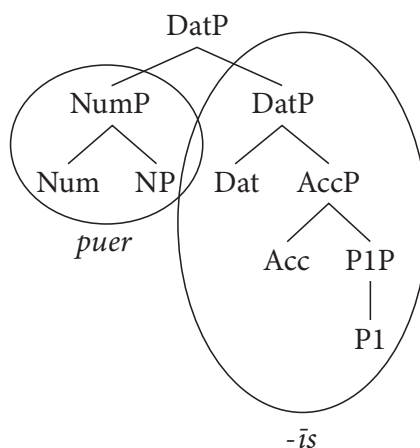


Figure 10.3 Structure for the form *puer-īs* ‘boys-DAT.PL’

Derivationally, after DAT was merged, the NumP in the specifier of AccP moves to Spec, DatP and the constituent [Dat [Acc [Pl]]] is realized as /īs/. This step raises the following question: what has happened with the lexicalization of the subconstituent [Acc [Pl]] as /ōs/? The idea is that the later lexicalization overrides the older lexicalization, which prevents the affix /os/ from arising in the dative case form.

Another important property of Nanosyntax is the lack of the notion of contextual allomorphy (understood as morphologically-conditioned allomorphy; the topic of phonological conditioning of allomorphy is outside the scope of this chapter; see Scheer 2016 for an influential discussion): there is only one way to expone any given syntactic structure and, thus, any difference in form found under the label of ‘contextual allomorphy’ implies exponence of different syntactic structures (see Caha et al. 2024; note, however, that the difference might be

purely configurational: [Dat [Gen Acc]] and [[Dat Gen] Acc] are different syntactic objects despite being composed of the same feature-heads). The exponence mechanism of Nanosyntax takes a constituent and matches it to a morphophonological representation: no context-sensitivity is possible. This is an important point of divergence between Nanosyntax and the main competing late-insertion syntactic approach to morphology, DM (see Halle and Marantz 1994; Bonet and Harbour 2012; Gouskova and Bobaljik 2020 for discussion of insertion in DM and see Caha 2018 for discussion of differences between DM and Nanosyntax with respect to insertion).

Taking stock, the main notions introduced in this section are that Nanosyntax is another member of the family of late-insertion syntactic approaches to morphology, that Nanosyntax employs mapping of whole phrases onto morphological exponents (Phrasal Lexicalization), and that Nanosyntax is committed to a bottom-up, derivational approach to lexicalization of syntactic structures, and achieves exponence of multiple feature heads together via syntactic movement and nothing else (Caha 2018). In the next section, we will tackle the question of the possibility of certain pseudo-ABA patterns given the basic Nanosyntactic commitments.

10.3 Nanosyntax and (pseudo-)ABA

This section discusses the derivation of certain pseudo-ABA patterns given the Nanosyntactic machinery outlined in the previous section. Firstly, we will discuss how Nanosyntax derives existing (and well discussed) cases of A–B–Ax patterns, taking the Malayalam pronominal paradigm as an example (Blix 2021, Middleton 2021). Then, I will discuss how Nanosyntax treats root suppletion and show why a straightforward idea for deriving three-cell portmanteau ABA patterns is untenable. Finally, building on discussion in the first two sections, I argue that the Nanosyntactic model is nevertheless able to derive ABA patterns, albeit in a more intricate fashion.

10.3.1 Nanosyntax and A–B–Ax patterns: backtracking or CLBs

The ban on ABA patterns in morphology is one of the most remarkable discoveries of morphological theory in recent years and is one that has started a fruitful research program of finding counterexamples and reanalyzing them in a way that does not lose the theoretical advancements triggered by the basic *ABA generalizations. This section discussed how Nanosyntax derives these patterns, using the Malayalam pronominal paradigm (Middleton 2021) as an example. The paradigm itself is given in Table 10.3. More precisely, Malayalam presents an A–B–Ax pattern

where the two cells of the paradigm are portmanteaus and the third cell is the first cell plus an additional affix (namely, a reduplication affix, assuming an item-and-arrangement approach to reduplication; see Marantz 1982; Raimy 1999 among others).

Table 10.3 The Malayalam pronominal paradigm (pronoun, diaphor, anaphor) (see Middleton 2021)

Pronoun	Diaphor	Anaphor
<i>avan</i> pronoun	<i>tan</i> diaphor	<i>avan-avan</i> pronoun-RED

This pattern can be derived in two ways. The first way to derive the A–B–Ax pattern is via ‘backtracking.’ Recall that the LA specifies a step of going to the previous cycle of the algorithm, which is often referred to as ‘backtracking.’ Backtracking at work can be seen in the derivation with the lexicon specified in Figure 10.4. Here, I assume Middleton’s containment structure for pronouns, diaphors, and anaphors: [Anaphor [Diaphor [Pronominal XP]]].

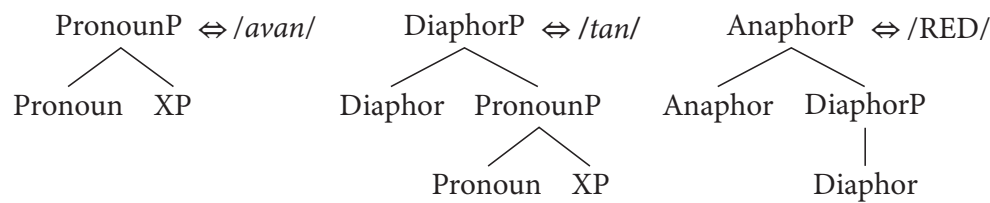


Figure 10.4 Lexicon for Malayalam pronominal paradigm

Consider the derivation (assuming, again, the Anaphor>Diaphor>Pronoun *fseq*) presented below. First, we merge Pronoun head to XP, which matches the lexical entry for /*avan*/ (Figure 10.5). After we merge Diaphor head to the pronoun structure, the resulting phrase matches the lexical entry for /*tan*/ (Figure 10.5).

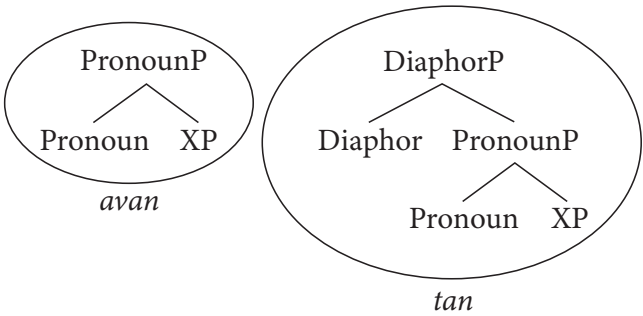


Figure 10.5 Lexicalization of PronounP and DiaphorP

After we merge the Anaphor head to the DiaphorP structure, however, the resulting structure cannot be lexicalized (Figure 10.6). Moving the DiaphorP structure to Spec, AnaphorP also does not result in a lexicalizable structure (notational convention is to mark a non-lexicalizable structure by exclamation marks). According to the LA, one has to return to the previous cycle and move the pronoun structure to Spec, DiaphorP (Figure 10.6). The resulting structure can be lexicalized as /*avan*-RED/ (since this is an intermediate lexicalization step in lexicalization of the anaphor, it is not a problem that the diaphor structure is realized as *avan-avan* and not *tan*).

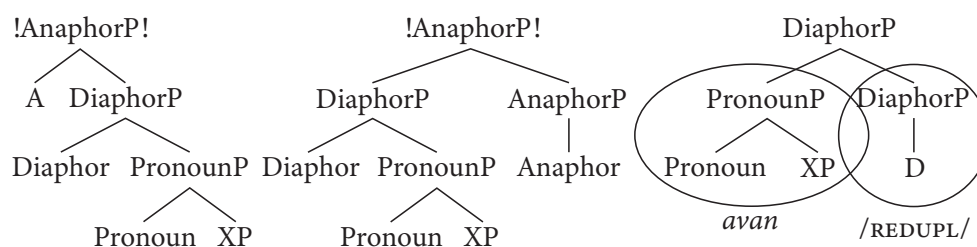


Figure 10.6 The current cycle does not result in a licit lexicalization: re-lexicalization via backtracking

Then, when we merge the Anaphor head, although the resulting structure cannot be lexicalized, the next step of the algorithm (move the pronoun structure to Spec, AnaphorP) results in a structure, the substructures of which match the lexical entries of /*avan*/ and /RED/, deriving the form *avan-avan* for the anaphor, both steps are shown in Figure 10.7.

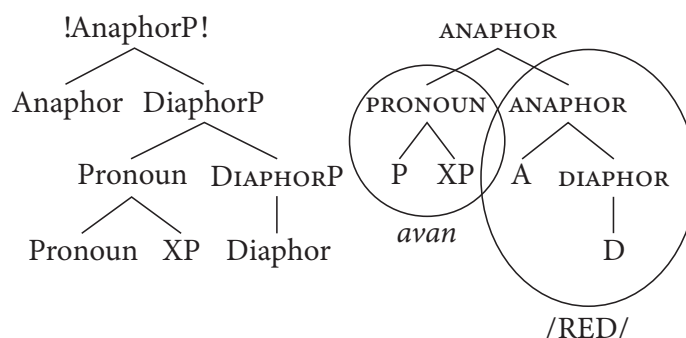


Figure 10.7 Deriving the anaphor form *avan-avan* (*avan*-/RED/)

This case study shows that the Nanosyntactic LA and its representational means are expressive enough to derive pseudo-ABA patterns of the A-B-Ax sort. However, Blix (2021) presents an alternative derivation for pseudo-ABA patterns that does not involve backtracking. His idea is to push the notion of phrasal lexicalization to its limits and introduce lexical entries with movement (thus having CLBs of the sort seen in Figure 10.7 included in the lexical entries). His proposal for Malayalam lexical entries is given in Figure 10.8. The lexical entry for diaphors contains

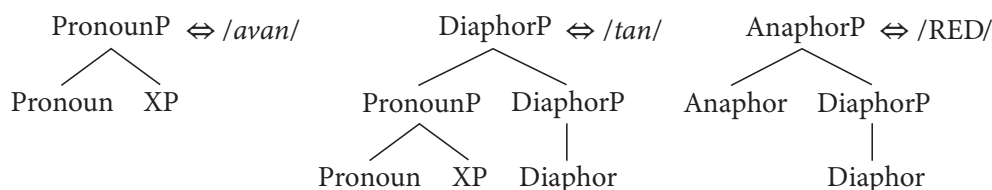


Figure 10.8 Blix's lexicon for Malayalam pronominal paradigm

a CLBs: the lexical entry can be matched to the whole structure after movement of the pronoun phrase to Spec, DiaphorP.

Consider the derivation with the provided lexicon. The first step would be to merge Pronoun head to XP and match the resulting structure with the lexical entry for *avan* (Figure 10.9). The next step is to merge Diaphor to the pronoun structure and, unlike the previous derivation, the resulting structure cannot be lexicalized. However, after movement of the pronoun structure to Spec, DiaphorP, the resulting structure can be matched with the lexical entry for /*tan*/. Similarly, after merging Anaphor head to DiaphorP, the resulting structure cannot be lexicalized, but after movement of Diaphor to Spec, AnaphorP the resulting structure's two substructures can be matched with the lexical entries for /*avan*/ and /*RED*/.

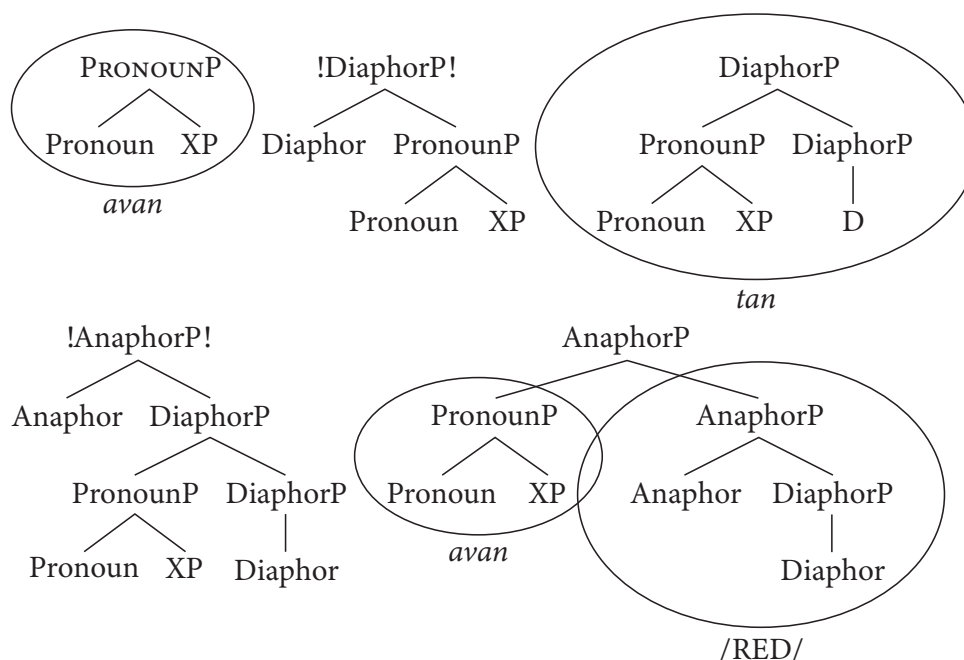


Figure 10.9 The derivation with complex left branches, Blix-style

The aim of this section was to show that the model of Nanosyntax presented in the previous section has enough expressive power to derive pseudo-ABA patterns of A–B–Ax type. Although this section has discussed ABA distribution of a functional item, the derivations presented in this section can be replicated for root

suppletion as well but this invites discussion of root suppletion in Nanosyntax, which is the topic of the next section.

10.3.2 Nanosyntax and root suppletion: pointers

Root suppletion is described as follows: the same abstract stem/root having different realizations in different cells of the paradigm. For example, English adjective *bad* has a suppletive form *worse*. This family of phenomena is rather problematic in the late-insertion syntactic approaches to morphology. It is often argued that the syntactic computation is ‘blind’ to the differences between different roots (given that the differences between roots appear to be either semantic, or phonological in nature, the sensitivity of syntax to the difference between roots appears to violate the commitments of generative grammar regarding modularity; see Vanden Wyngaerd, De Clercq, and Caha 2021 for discussion). The problem is the restriction of, say, *worse* to the comparative form of *bad*, given the supposed syntactic indistinctness of the syntactic root for *bad* and the syntactic root for *good*.

One of the existing solutions to this problem is to abandon the indistinctness of roots, as argued for by Harley (2014). However, alternatives exist. One of them is the Nanosyntactic solution to the problem, proposed in Vanden Wyngaerd, De Clercq, and Caha 2021. The solution has two parts. The first ingredient is the so-called pointer (see Figure 10.10). Assuming the split structure for comparatives (Caha, De Clercq, and Vanden Wyngaerd 2019), a Nanosyntactic lexicon for *bad* and *worse* is provided below. The restriction of *worse* to the comparative form of *bad* is achieved via stipulating in the lexical entry for *worse* that the structure can be realized as *worse* only if it is built on top of a structure previously lexicalized as *bad*.

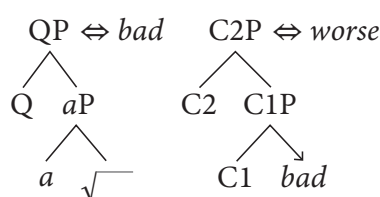


Figure 10.10 Pointers for root suppletion in Nanosyntax

The second part of the solution is the restriction of overriding previous lexicalization via the Faithfulness Principle. According to this principle, a lexicalization can override a previous lexicalization iff (a) the results of the lexicalizations are the same; (b) the lexical entry for the newer lexicalization contains a pointer to the older lexicalization. The Faithfulness Principle effectively restricts the ‘free choice’ of the root to the earliest stages of the derivation: the choice is free, but once it has been made, it is fixed.

(2) Faithfulness Principle

$/\alpha/$ can override a previous lexicalization $/\beta/$ iff

- a. $/\alpha/ = / \beta/$
- b. the lexical entry for $/\alpha/$ contains a pointer to $/\beta/$

The Faithfulness Principle rules out overriding given putative lexical entries such as those in Figure 10.11. Given the indistinctness of roots, the system without a variant of the Faithfulness Principle does not rule out realization of *aP* as *bad* and QP as *good*. The Faithfulness Principle, as said earlier, restricts the free choice: although the root can be realized via any root-containing lexical entry, the choice is set once made: override without pointers is rendered impossible by the Faithfulness Principle.

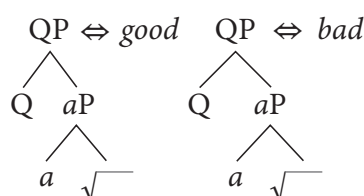


Figure 10.11 Two root-containing lexical entries

To show the principle at work in a more relevant example, let me show how the Faithfulness Principle rules out an ABA pattern encoded in the lexicon provided in Figure 10.12 (which is an attempt to derive a Malayam-style paradigm but without reduplication), which may seem as deriving an ABA pattern.

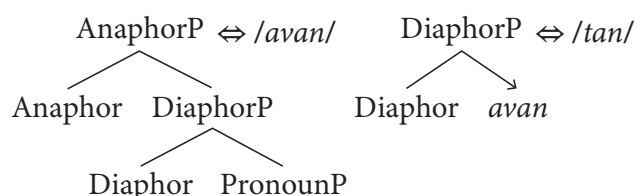


Figure 10.12 A candidate for an ABA derivation

The idea is that PronounP can only be matched to the lexical entry for *avan*, DiaphorP matches *tan* (since it consists of Diaphor on top of a constituent previously lexicalized as *avan*), and AnaphorP matches *avan* again, potentially deriving an ABA suppletion pattern. However, it is evident that the Faithfulness Principle rules out such derivations. Overriding the lexicalization of Diaphor as *tan* by lexicalizing AnaphorP as *avan* is illicit since the lexical entry for *avan* does not contain a pointer to *tan*. In general, it seems that pure ABA patterns of suppletion are illicit since they would require two lexical entries containing pointers to each other, a useless lexicon since none of the lexical entries would ever be used.

This argument appears to be a welcome result: pure ABA patterns of suppletion (where all three members of the paradigm are portmanteaus) are out. However, in the next section, I aim to argue that such patterns can nevertheless be derived by the Nanosyntactic model under consideration in this chapter, albeit that the derivation requires more intricate lexical entries.

10.3.3 Nanosyntax and ABA patterns: pointers to CLBs

Consider the lexicon in Figure 10.13. In this section, I wish to argue that the lexicon in Figure 10.13, which is allowed by the representational rules of Nanosyntax, suffices to derive an ABA root suppletion pattern where all three cells of the paradigm are portmanteaus.

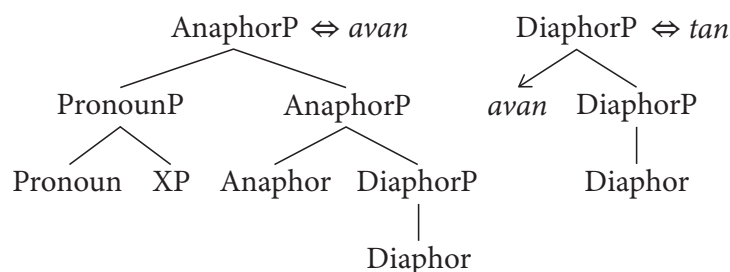


Figure 10.13 Lexicon for an ABA suppletion pattern (in a pseudo-Malayam)

The derivation is rather similar to the derivation of the Malayalam pronominal paradigm as presented in Blix (2021). The first step is to merge Pronoun to the XP, shown in Figure 10.14. Given the Superset Principle, the lexical entry for *avan* matches PronounP, given that PronounP is a subconstituent of the syntactic structure in the lexical entry for *avan*. For the clarity of pointer application, I will retain the lexicalization in the next steps of the derivation by employing a pair <SYN, PHON> notation.

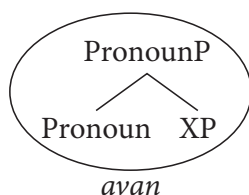


Figure 10.14 Merging Pronoun to the XP

Then, Diaphor head is merged. DiaphorP in Figure 10.15 cannot be lexicalized: it is not a subconstituent of the lexical entry for *avan* and it also does not match to the lexical entry for *tan*, given that it does not contain a unary branching DiaphorP. Since it is not lexicalized, the next step in the LA is to move the specifier of PronounP to Spec, DiaphorP. However, there is no specifier of PronounP,

which leads to the next step of the algorithm, namely, movement of the whole PronounP to Spec, DiaphorP. The resulting structure matches to the L-tree for *tan*: it contains a unary branching DiaphorP whose (left) sister has been lexicalized as *avan* in the previous cycle of lexicalization.

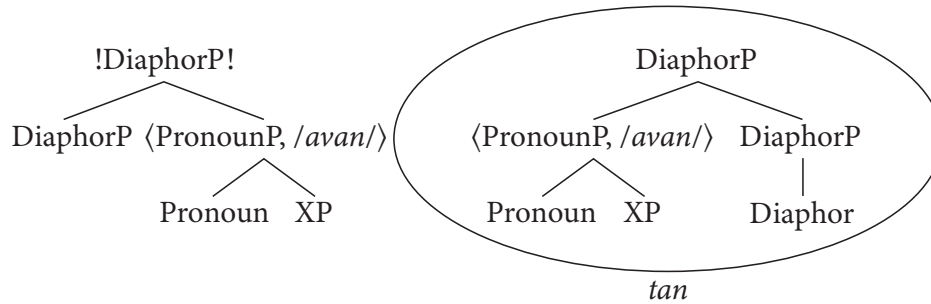


Figure 10.15 Lexicalizing DiaphorP

The crucial step is merging Anaphor head to DiaphorP. The immediate resulting structure does not match to either of the lexical entries provided in and, thus, the next step in the LA is to move the specifier of DiaphorP to Spec, AnaphorP, which results in the structure where PronounP is in the Spec, AnaphorP, which is a perfect match to the lexical entry for */avan/* (as in Figure 10.16).

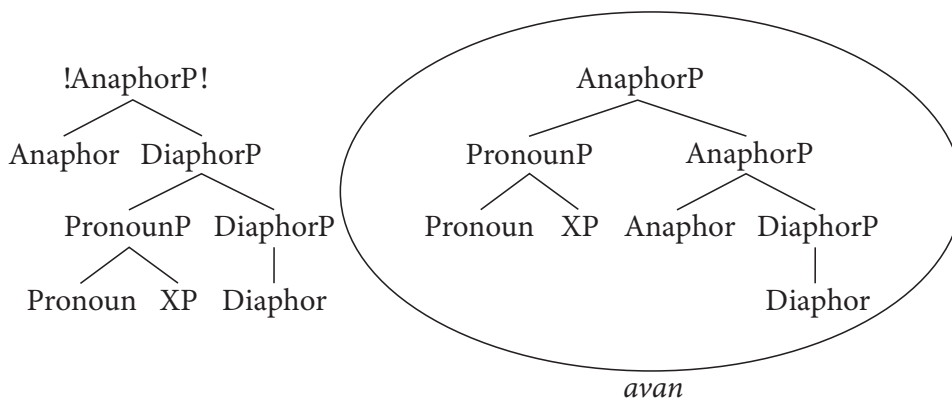


Figure 10.16 Lexicalizing AnaphorP

The resulting paradigm is provided in Table 10.4 and it is clear that the derivation outlined above results in an ABA root suppletion pattern. There are three cells of the paradigm, featurally represented as [Pronoun], [Pronoun+Diaphor], [Pronoun+Diaphor+Anaphor], expounded as *avan*, *tan*, and *avan*, respectively. This is an ABA root suppletion pattern, and it can be derived in Nanosyntax.

It is worth discussing why this derivation does not violate the Faithfulness Principle, unlike the putative ABA derivation discussed in the previous section. The idea is that, after movement of PronounP to Spec, AnaphorP, there is no constituent realized as */tan/* at all, so, no unfaithful override occurs. Note that the derivation outlined above complies with the Faithfulness Principle on any

Table 10.4 The resulting paradigm of pseudo-Malayalam

[Pronoun]	[Diaphor + Pronoun]	[Anaphor + Diaphor + Pronoun]
<i>avan</i>	<i>tan</i>	<i>avan</i>

understanding of the Cyclic Override property of the Nanosyntactic LA. If the derivation ‘remembers’ that PronounP has been lexicalized as /*avan*/ (see left side of Figure 10.17), the lexicalization of AnaphorP as /*avan*/ then overrides the previous lexicalization of PronounP as /*avan*/, which complies with the Faithfulness Principle. Alternatively, if the derivation ‘deletes’ all previous lexicalizations (see right side of Figure 10.17), then no override happens in the lexicalization of AnaphorP as /*avan*/.

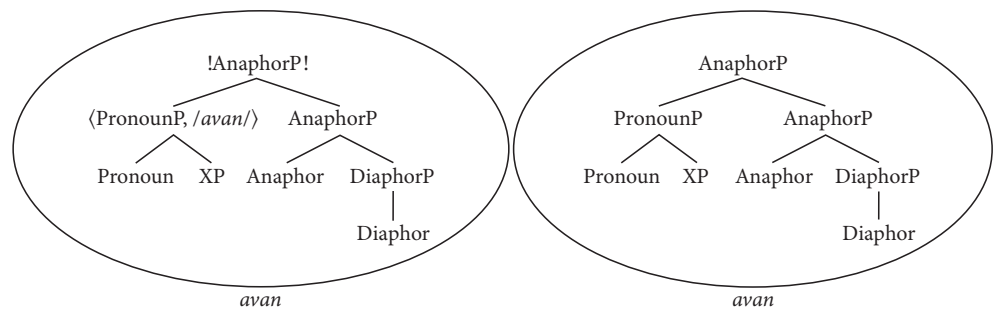


Figure 10.17 Two lexicalization options

The result raises the question of whether this theoretical possibility is empirically attested, or whether the theory should be further restricted to disallow this possibility. One could, for example, aim for a ban on pointers to specifiers, which would rule out such lexicons as the one discussed in this section. However, I wish to make the point that the prediction is welcome and supported by crosslinguistic data. The next section presents a puzzling pattern in Russian that presents a case of the ABA root suppletion pattern just like the ones argued to be possible in this section, given the Nanosyntactic machinery. If my characterization of the Russian case is correct, the derivation discussed in this section is then not an example of overgeneration, as one could potentially argue.

10.4 Support from Russian

This section introduces the data of interaction between suppletive adjectives and analytic degree morphology in Russian which may present a case of an ABA portmanteau pattern. The section is structured as follows: first, I present the pattern. Then, I discuss two possible analyses, one of which involves an ABA

paradigm—the difference between two analyses lies in a currently unsolved problem of optional ‘intensifying’ prefix *nai-* in Russian superlatives. Based on the distributional properties of *nai-*, I argue that the data does present a three-cell portmanteau ABA pattern, thus supporting the theoretical prediction highlighted in the previous section.

10.4.1 The pattern and analytical options

The data comes from a puzzling interaction between Russian analytic superlatives (*samyj*-superlatives) and suppletive adjectives. While most adjectives form *samyj*-superlatives by combining *samyj* with their positive form (see Table 10.5), some Russian speakers do not allow that for adjectives with suppletive superlatives, preferring combining *samyj* with the suppletive superlative instead (see Table 10.6).

Table 10.5 Analytic superlative form of regular adjectives

Positive	Superlative	Analytic superlative
<i>glup-yj</i> dumb-AGR ‘dumb’	<i>glup-ej-š-ij</i> dumb-CMPR-SPRL-AGR ‘dumbest’	<i>samyj glup-yj</i> most dumb-AGR ‘most dumb’

Table 10.6 Analytic superlative form of suppletive adjectives (for some Russian idiolects)

Positive	Superlative	Analytic superlative
<i>xoroš-ij</i> good-AGR ‘good’	<i>lučš-ij</i> best-AGR ‘best’	<i>samyj lučš-ij</i> most best-AGR ‘most good’

It should be noted that, to an extent, the pattern replicates for Russian analytic comparatives formed using *bole* ‘more’ (see Matushansky (2002) for additional discussion and see Kosheleva (2016) for discussion of the factors governing the choice of the comparative form). Although the judgments are less clear, I make a simplifying assumption and equate the comparative and superlative patterns to each other, as presented in the Tables 10.7 and 10.8.

Table 10.7 Analytic comparative form of regular adjectives

Positive	Comparative	Analytic comparative
<i>glup-yj</i> dumb-AGR ‘dumb’	<i>glup-ej-e</i> dumb-CMPR-AGR ‘dumber’	<i>samyjglup-yj</i> more dumb-AGR ‘more dumb’

Table 10.8 Analytic comparative form of suppletive adjectives

Positive	Comparative	Analytic comparative
<i>xoroš-ij</i> good-AGR ‘good’	<i>lučš-e</i> better-AGR ‘better’	<i>bolee lučš-ij</i> more better-AGR ‘more better’

Regardless of the theoretical claim of Section 10.2, the pattern of interaction between (suppletive) adjectives and analytic degree forms is interesting on its own and the secondary goal of this chapter is thus to provide a Nanosyntactic analysis of the pattern.¹ The main problem posed by the data is that the form of the lexical adjectives found in their analytic forms is not consistently syncretic with either the positive form (as is the case with regular adjectives) or the comparative/superlative form (as is the case with suppletive adjectives). Therefore, the analytic form of adjectives has to be its own paradigm cell, which is sometimes syncretic with the positive form and sometimes syncretic with the comparative/superlative form (this line of reasoning is the same that is employed by, for example, Caha (2009) to motivate accusative case in Estonian and Finnish as its own paradigm cell, despite it never having a special exponent of its own). Assuming that there is a special ‘analytic’ paradigm cell in the degree allomorphs of Russian adjectives, there are three possible configurations of the three-cell paradigm consisting of the positive form, the comparative form, and the analytic form, outlined in Tables 10.9–10.11.

In the next section, I aim to establish that the attractive, ABA-avoiding option in Table 10.10 is not viable, provided other observations about Russian degree morphology (building on the discussion of the so-called zero-comparatives in Kasenov (2023)). In essence, the main result of this section is a proof of concept regarding the theoretical prediction outlined in Section 10.3.

Table 10.9 Analysis with AAB for regular adjectives and ABA for suppletive adjectives

Analytic	Positive	Comparative
<i>glup-</i>		<i>glup-ej-</i>
<i>lučš-</i>	<i>xoroš</i>	<i>lučš-</i>

10.4.2 Analytic forms of suppletive adjectives present an ABA pattern

As stated earlier, the best analytical outcome would be the paradigm in Table 10.10: no ABA distribution of allomorphs whatsoever and the pattern is thus rendered

¹ One question that I abstract away from is how prefix formation works in Nanosyntax. For the current purposes, the following suffices: when taken together, the prefixal part and the non-prefixal part should realize all the *fseq* required. The exact implementation is not crucial, but see Starke (2018) for a classic discussion.

Table 10.10 Analysis with AAB for regular adjectives and ABB for suppletive adjectives

Positive	Analytic	Comparative
<i>glup-</i>		<i>glup-ej-</i>
<i>xoroš-</i>	<i>lučš-</i>	

Table 10.11 Analysis with ABA for regular adjectives and ABB for suppletive adjectives

Positive	Comparative	Analytic
<i>glup-</i>	<i>glup-ej-</i>	<i>glup-</i>
<i>xoroš-</i>	<i>lučš-</i>	

unproblematic. Let us then attempt to work out a formal model that would provide the paradigm in Table 10.10. One problem is that it assumes that there is some structure which (a) is contained in the comparative form; (b) contains the positive form. Although the structural proposal for comparatives in, say, Bobaljik (2012) does not allow such a structure to even exist, the more finely articulated structural proposals, such as the one given in Caha, De Clercq, and Vanden Wyngaerd (2019), do. Nanosyntactic work on degree morphology (of which the study in Caha, De Clercq, and Vanden Wyngaerd (2019) is an exemplar) argues that the functional sequence of feature-heads employed is *a*-Q-C1-C2-S1-S2. The featural makeup of the main three forms is provided in (3).

(3) Nanosyntactic structure for

- a. positive form:] *a*] Q]
- b. comparative form:] *a*] Q] C1] C2]
- c. superlative form:] *a*] Q] C1] C2] S1] S2]

Given this background, an implementation of the paradigm in Table 10.10 would be that the analytic form is of size C1P, while the positive form is of size QP and the comparative form is of size C2P. From the Nanosyntactic, cyclic lexicalization point of view, this means that the stem of regular adjectives can realize C1P, suggesting the lexicalization for the regular adjectives outlined in Table 10.12.

There is language-internal evidence against such an analysis that comes from a curious class of adjectives which do not show that comparative *-ej-* affix in their comparative form, but do show it in the superlative form, such as *krutoj* ‘cool,’ the paradigm for which is provided in Table 10.13. I should note that the *krut-kruč* alternation is a case of a morphophonological process called palatalization; see Blumenfeld (2003) for an overview.

The paradigm in Table 10.13 is puzzling because the superlative form *krut-ej-š-* contains the ‘comparative’ form *krut-ej-* in the same way as the superlative form *glup-ej-š-* contains the comparative form *glup-ej-*. The issue is, then, that there

Table 10.12 Lexicalization for regular adjectives (*version 1*)

aP	Q	C1	C2	S1	S2
glup					
glup					
glup-					
glup			ej		
glup			ej	-š-	
glup			ej	-š-	

Table 10.13 The degree paradigm of the adjective *krut-oj* ‘cool’

Positive	Comparative	Superlative
<i>krut-oj</i> cool-AGR ‘cool’	<i>kruč-e</i> cool-CMPR ‘cooler’	<i>krut-ej-š-ij</i> cool-CMPR-SPRL-AGR ‘coolest’

is no such comparative form as *krut-ej-* in Russian: **krut-ej-e* ‘cool-CMPR-AGR’ is an illicit word-form, while *kruč-e* ‘cool.CMPR-AGR’ is. Therefore, the lexicalization process for the superlative form of the adjectives like *krutoj* ‘cool’ will require re-lexicalization of the comparative part of the structure, in order for the comparative affix *-ej-* to arise in the superlative form while being absent from the comparative form.

The core idea for the analysis presented in Kasenov (2023) is that the lexicalization of the superlative part of the *fseq* for zero-comparative adjectives triggers backtracking (and thus, re-lexicalization). There are, essentially, two ways to do so: the first way is to have *overlap* between the lexical entry for the superlative affix and the comparative part of *fseq* (for example, if the lexical entry for */-š/* has C2 as its ‘foot,’ the C1P-part of the structure has to move to Spec, S1P in order for lexicalization to be possible). This option is fleshed out in the lexicalization table below (Table 10.14) (this option corresponds to the analysis in Kasenov 2023)—note that the lexical entry for *krut-* is thus a CLB entry in order to derive the A–Ax–A pattern between QP, C1P, and C2P (see Blix 2022 for an example of such derivation for Kipsigis singulatives).

The second way to trigger re-lexicalization, however, is to have *a gap* between the superlative affix’s lexical entry and the comparative part of *fseq*.² If the lexical

² I thank Pavel Caha for bringing this possible analysis to my attention.

Table 10.14 *Overlap*-lexicalization for zero-comparatives adjectives

QP	C1	C2	S1	S2
krut-				
krut-	-ej-			
krut-				
krut	-ej-	-š-		
krut	-ej-	-š-		

entry for the stem of zero-comparative cannot expone the S1 head and the superlative affix cannot either, the LA will trigger re-lexicalization. Under the assumption that the comparative affix *-ej-* can realize the S1, the zero-comparative pattern becomes possible to analyze, as shown by the lexicalization configuration in Table 10.15.

Table 10.15 *Gap*-lexicalization for zero-comparatives adjectives

QP	C1	C2	S1	S2
krut-				
krut-				
krut-				
krut		-ej-		
krut		-ej-		-š-

The choice between the two analyses is directly relevant to the viability of an ABA-avoiding analysis of the interaction between suppletive adjectives and analytic degree morphology: if */-ej-/* arises in the lexicalization of the C1P for zero-comparatives, then we expect */-ej-/* to arise in the analytic form as well, contrary to the fact (*boleē krutoj* ‘more cool’ vs **boleē krut-ej* ‘more cool-CMPR’). As far as I am aware, only one corner of Russian degree morphology appears to present an empirical testing ground for the two proposals with respect to the lexical entry of the superlative affix */-š/*: the optional ‘intensifying’ prefix *nai-*.

- (4) *glup-ej-š-ij* ‘stupid-CMPR-SPRL-AGR’
nai-glup-ej-š-ij ‘nai-stupid-CMPR-SPRL-AGR’

There are two core distributional facts about the prefix *nai-* which appear to support treating it as an exponent of degree morphology: (i) it never appears outside of degree forms and has no ‘intensifying’ function elsewhere; (ii) it can be used to

form superlative forms of comparative adverbials (a peculiar fact is that this strategy seems to be restricted to adverbials of suppletive adjectives—accounting for this observation is outside the scope of this chapter).

- (5) *bol'-ej-e* 'more-CMPR-AGR' — *nai-bol'-ej-e* 'SPRL-more-CMPR-AGR'
men-ej-e 'less-CMPR-AGR' — *nai-men-ej-e* 'SPRL-less-CMPR-AGR'
lučš-e 'better-AGR' — *nai-lučš-e* 'SPRL-better-AGR'
xuž-e 'worse-AGR' — *nai-xuž-e* 'SPRL-worse-AGR'

The conclusion I draw from (i)–(ii) is that the lexical entry for *nai-* should include some part of the superlative degree structure. This conclusion is incompatible with a *gap*-lexicalization analysis, according to which the *-š-* suffix realizes only the C2 head—combination of *nai-* prefix with the *-š-* suffix is predicted to be impossible (see Starke 2018 on the discussion of structural conditions for prefix formation). Therefore, only the *overlap*-lexicalization is a viable analysis of the zero-comparative paradigms. Therefore, the ABA-avoiding lexicalization in Table 10.10 is impossible—claiming that the 'analytic form' is of size C1P makes the wrong prediction regarding the analytic form of the zero-comparative adjectives like *krutoj* 'cool.'

10.4.3 Consequences of ABA

The preceding discussion established that the data present an ABA pattern and should be analyzed by either of the paradigms presented in Table 10.16 and 10.17 (repeated from Tables 10.9 and 10.11). For concreteness, I assume the paradigm Table 10.16, which postulates an ABA pattern with suppletive adjectives (it also seems that the ABA pattern with regular adjectives will turn out to be problematic, once the superlatives are taken into account but I do not wish to focus on that).

Table 10.16 AAB with regular adjectives, ABA with suppletive adjectives

Analytic	Positive	Comparative
<i>glup-</i>		<i>glup-ej-</i>
<i>lučš-</i>	<i>xoroš</i>	<i>lučš-</i>

Assuming the results of Nanosyntactic work on degree morphology, the paradigm can be recast using the functional sequence in a way provided in Table 10.18, with *aP* corresponding to the 'analytic' root, *QP* the positive one, and *C1P* and *C2P* the comparative root. The objective is now to provide a Nanosyntactic lexicon such that it generates the pattern *lučš–xoroš–lučš*.

Table 10.17 ABA with regular adjectives, ABB with suppletive adjectives

Positive	Comparative	Analytic
<i>glup-</i>	<i>glup-ej-</i>	<i>glup-</i>
<i>xoroš-</i>	<i>lučš-</i>	

Table 10.18 The resulting paradigm for regular and suppletive adjectives

<i>aP</i>	QP	C1P	C2P
<i>glup-</i>	<i>glup-</i>	<i>glup-ej</i>	<i>glup-ej</i>
<i>lučš-</i>	<i>xoroš-</i>	<i>lučš-</i>	<i>lučš-</i>

In light of the discussion in the previous section, the lexicon that generates such a pattern is provided in Figure 10.18. Abstracting away from the C2 head, the lexical entries are essentially the same as those presented in (18) in the previous section as a pure representational possibility that could generate an ABA portmanteau pattern—hence it has been already shown that the lexicon is enough to generate an ABA pattern presented by the Russian data.

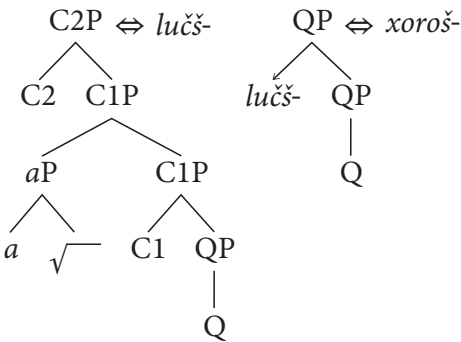


Figure 10.18 ABA-generating lexicon for the pattern

If my characterization of the pattern as an example of ABA is correct, then it presents a case of the pattern, which should be analyzed using the movement-containing suppletion schemata proposed in Section 10.3.

An important caveat to the picture presented earlier is that there is inter-speaker variation with respect to the interaction between suppletive adjectives and analytic degree morphology. Some Russian speakers behave as one would expect, and combine the positive form of such adjectives with the analytic comparative/superlative form markers (see Table 10.19 and Table 10.20).

The relevance of the existence of such speakers concerns the putative characterization of the ban on ABA patterns as the result of extra-grammatical

Table 10.19 Analytic comparatives of suppletive adjectives in other Russian idiolects

Positive	Comparative	Analytic comparative
<i>xoroš-ij</i> good-AGR 'good'	<i>lučš-e</i> better-AGR 'better'	<i>bolee xoroš-ij</i> more good-AGR 'more good'

Table 10.20 Analytic superlatives of suppletive adjectives in other Russian idiolects

Positive	Superlative	Analytic superlative
<i>xoroš-ij</i> good-AGR 'good'	<i>lučš-ij</i> best-AGR best	<i>samyj xoroš-ij</i> most good-AGR 'most good'

forces (Andersson 2018). It may be that positing lexical entries in the style of the movement-containing suppletion schemata requires much more data in support of it than more regular suppletion lexical entries, without movement, in which the ban on ABA falls out of the formal computational system. The existence of inter-speaker variation may be then regarded as evidence for ABA patterns being hard to learn due to the complexity of the movement-containing suppletion schemata.

10.5 Conclusion

This chapter has discussed the limits of the current Nanosyntactic model with respect to its generative power regarding (pseudo-)ABA phenomena. I have argued that existing tools in the Nanosyntactic machinery allows ABA suppletion patterns where all three cells are portmanteau and have provided evidence from Russian adjectival morphology that such patterns are attested. Even though I am optimistic about the conclusion not being fatal for the Nanosyntactic model, someone else might be—there are ways to block the derivation presented in the chapter by either constraining the power of pointers (a conclusion argued for by Vanden Wyngaerd (2018)) or by modifying the Faithfulness Principle. However, I also believe that the Nanosyntactic system is nevertheless constrained with respect to ABA phenomena, since the pattern discussed in this work still conforms to the forming consensus that true portmanteaus are beyond all cases of seemingly

ABA-ish distribution of morphological phenomena (see the conclusion in Davis 2021). Finally, the ability of the Nanosyntactic computational system to derive ABA phenomena may lead the research agenda to look into ‘third-factor’ explanations of the rarity of ABA patterns in morphosyntactic typology (as argued for by Andersson 2018: 10).

References

- Andersson, Samuel. 2018. “(*)ABA in Germanic Verbs.” *Glossa: A Journal of General Linguistics* 3(1).
- Arregi, Karlos and Andrew Nevins. 2012. *Morphotactics: Basque Auxiliaries and the Structure of Spellout*. Dordrecht: Springer.
- Baunaz, Lena and Eric Lander. 2018. “The Basics of Nanosyntax.” In *Exploring Nanosyntax*, ed. Lena Baunaz, Liliane Haegeman, Karen De Clercq, and Eric Lander. Oxford: Oxford University Press, 3–56.
- Blix, Hagen. 2021. “Phrasal Spellout and Partial Overwrite: On an Alternative to Backtracking.” *Glossa: A Journal of General Linguistics* 6(1), 62: 1–17.
- Blix, H. 2022. “Interface Legibility and Nominal Classification: A Nanosyntactic Account of Kipsigis Singulatives.” *Glossa: A Journal of General Linguistics* 7(1), 1–46.
- Blumenfeld, Lev. 2003. “Russian Palatalization and Stratal OT: Morphology and [Back].” In *Proceedings of the Annual Workshop on Formal Approaches to Slavic Linguistics*, vol. 11, 141–58.
- Bobaljik, Jonathan David. 2000. “The Ins and Outs of Contextual Allomorphy.” *University of Maryland Working Papers in Linguistics* 10: 35–71.
- Bobaljik, Jonathan David. 2012. *Universals in Comparative Morphology: Suppletion, Superlatives, and the Structure of Words*. Cambridge, MA: MIT Press.
- Bobaljik, Jonathan David and Uli Sauerland. 2018. “ABA and the Combinatorics of Morphological Features.” *Glossa: A Journal of General Linguistics* 3(1), 15: 1–34.
- Boeckx, Cedric. 2014. *Elementary Syntactic Structures: Prospects of a Feature-Free Syntax*. Cambridge, MA: Cambridge University Press, vol. 144.
- Bonet, Eullia and Daniel Harbour. 2012. “Contextual Allomorphy.” In *The Morphology and Phonology of Exponence*, ed. Jochen Trommer. Oxford: Oxford University Press, 195–235.
- Caha, Pavel. 2009. “The Nanosyntax of Case.” Universitetet i Tromsø dissertation.
- Caha, Pavel. 2017. “Suppletion and Morpheme Order: Are Words Special?” *Journal of Linguistics* 53(4): 865–96.
- Caha, Pavel. 2018. “Notes on Insertion in Distributed Morphology and Nanosyntax.” In *Exploring Nanosyntax*, ed. Lena Baunaz, Liliane Haegeman, Karen De Clercq, and Eric Lander. Oxford: Oxford University Press, 57–87.
- Caha, Pavel. 2021. “Modeling Declensions without Declension Features: The Case of Russian.” *Acta Linguistica Academica* 68(4): 385–425.
- Caha, Pavel. 2024. “Root and Stem Allomorphy without Multiple Exponence: The Case of Special Nominatives.” *Morphology*, 1–45.
- Caha, Pavel, Karen De Clercq, and Guido Vanden Wyngaerd. 2019. “The Fine Structure of the Comparative.” *Studia Linguistica* 73(3): 470–521.

- Caha, Pavel, Karen De Clercq, and Guido Vanden Wyngaerd. 2024. "Allomorphy without Context Specification: A Case Study of Czech Adjectival Stems." *Morphology*, 1–44.
- Davis, Colin. 2021. "Case-Sensitive Plural Suppletion in Barguzin Buryat: On Case Containment, Suppletion Typology, and Competition in Morphology." *Glossa: A Journal of General Linguistics* 6(1): 1–26.
- Gouskova, Maria and Jonathan Bobaljik. 2020. "Allomorphy and Vocabulary Insertion." *Submitted to Handbook of Distributed Morphology*. <https://ling.auf.net/lingbuzz/005388>.
- Halle, Morris and Alec Marantz. 1994. "Some Key Features of Distributed Morphology." *MIT Working Papers in Linguistics* 21(275): 88.
- Harley, Heidi. 2014. "On the Identity of Roots." *Theoretical Linguistics* 40(3–4): 225–76.
- Hewett, Matthew. 2023. "Allomorphy in Semitic Discontinuous Agreement: Evidence for a Modular Approach to Postsyntax." *Natural Language & Linguistic Theory* 41(3): 1091–145.
- Kasenov, Daniar. 2023. "ABA in Russian Adjectives, Subextraction, and Nanosyntax." To appear in *Advances in Slavic Formal Linguistics 2022*. lingbuzz/007496.
- Kayne, Richard. 2005. *Movement and Silence*. Oxford: Oxford University Press.
- Kosheleva, Daria. 2016. "Rival Forms of Comparatives in Russian." *Poljarnyj vestnik: Norwegian Journal of Slavic Studies* 19: 42–50.
- Lander, Eric and Liliane Haegeman. 2018. "Syncretism and Containment in Spatial Deixis." In *Exploring Nanosyntax*, ed. Lena Baunaz, Liliane Haegeman, Karen De Clercq, and Eric Lander. Oxford: Oxford University Press, 116–48.
- Marantz, Alec. 1982. "Re Reduplication." *Linguistic Inquiry* 13(3): 435–82.
- Matushansky, Ora. 2002. "More of a Good Thing: Russian Synthetic and Analytic Comparatives." In *Formal Approaches to Slavic Linguistics 10: The Second Ann Arbor Meeting 2001*, ed. Toman Jindřich. Ann Arbor: Michigan Slavic Publications.
- Middleton, Jane. 2021. "Pseudo-ABA Patterns in Pronominal Morphology." *Morphology* 31(4): 329–54.
- Raimy, Eric Stephan. 1999. "Representing Reduplication." University of Delaware dissertation.
- Scheer, Tobias. 2016. "Melody-Free Syntax and Phonologically Conditioned Allomorphy." *Morphology* 26: 341–78.
- Starke, Michal. 2009. "Nanosyntax: A Short Primer to a New Approach to Language." *Nordlyd* 36(1): 1–6.
- Starke, Michal. 2018. "Complex Left Branches, Spellout, and Prefixes." In *Exploring Nanosyntax*, ed. Lena Baunaz, Liliane Haegeman, Karen De Clercq, and Eric Lander. Oxford: Oxford University Press, 57–87.
- Sudo, Yasutada and Andrew Ira Nevins. 2022. "No ABA Patterns with Fractionals." In *For Hagit: A Celebration*, ed. Linnaea Stockall, Luisa Martí, David Adger, Isabelle Roy, and Sarah Ouwayda. *Qmul Occasional Papers in Linguistics* 47, 1–10.
- Taraldsen, Knut Tarald. 2019. "Nanosyntax and Syncretism in Multidimensional Paradigms." *Linguistics Vanguard* 5(1): 20180058.
- Vanden Wyngaerd, Guido. 2018. "The Feature Structure of Pronouns: A Probe into Multidimensional Paradigms." In *Exploring Nanosyntax*, ed. Lena Baunaz, Karen De Clercq, Liliane Haegeman, and Eric Lander. Oxford: Oxford University Press, 277–304.
- Vanden Wyngaerd, Guido, Karen De Clercq, and Pavel Caha. 2021. "Late Insertion and Root Suppletion." *Revista Virtual de Estudos da Linguagem—ReVEL* 19 (18) 1–45.